OPEN

Assessing Veterans With Gulf War Illness for Prodromal Parkinson's Disease Risk With the REM Behavioral Sleep Disorder Questionnaire

Linda L. Chao, PhD, Tamara Stiep, MD, and José Rafael P. Zuzuárregui, MD

Objective: To assess deployed Gulf War (GW) veterans for prodromal Parkinson's Disease (PD) using the REM Sleep Behavioral Disorder (RBD) Screening Questionnaire (RBDSQ). **Methods:** GW veterans (n = 165) were assessed for nonmotor PD symptoms with the RBDSQ, Non-Motor Symptoms Scale, and Scales for Outcomes in Parkinson's Disease–Autonomic. Some veterans were also assessed with the University of Pennsylvania Smell Identification Test (n = 120), neurological examinations (n = 41), and 3T magnetic resonance imaging (n = 16). **Results:** Eighty percent of veterans with PD and 73% of veterans without PD had probable RBD (ie, RBDSQ ≥5). Exploratory analysis conducted for hypothesis-generating purposes revealed that RBDSQ scores were positively associated with GW deployment–related exposures and inversely associated with brainstem volume. **Conclusions:** These findings suggest that veterans with GWI and high levels of deployment-related exposures are at risk for prodromal PD, possibly through RBD as a mediator.

Keywords: Gulf War, Gulf War illness, Parkinson's disease, REM behavioral sleep disorder, brainstem

LEARNING OUTCOMES

- Describe rates of probable RBD, a marker of prodromal PD, in different subgroups of deployed GW veterans
- Understand the relationship between GW deployment–related exposures and risk for RBD and prodromal PD in GW veterans
- Understand the relationship between brainstem volume, risk for RBD, and prodromal PD in deployed GW veterans

Despite its brevity (August 2, 1990–February 29, 1991), the Gulf War (GW) was remarkable for the number of chemical exposures experienced by troops. For example, there was smoke from hundreds of burning oil well fires, extensive pesticide use, low levels of nerve agent released from the destruction of Iraqi facilities, and vaccines and prophylactic medications used to protect troops from potential exposure to biological and chemical warfare agents.[1] Gulf War illness (GWI) is a multisymptom condition estimated to affect 25%–32% of deployed GW veterans.[2] The US Institute (now National Academy) of Medicine's Committee on Gulf War and Health has concluded that there is sufficient evidence linking GWI to military service in the 1991 Gulf War.[3] In the decades since the end of the Gulf War, research on the etiology of GWI has implicated some of the deployment-related chemicals.[1,4–7] Notably, some of the chemicals present in the GW milieu (eg, organophosphorus pesticides, organochlorines, and permethrin) have been linked to increased risk for Parkinson's disease (PD),[8] the second most common age-related neurodegenerative disease.[9]

Although the clinical signs most commonly associated with PD are motor symptoms (eg, tremor, rigidity, and bradykinesia), PD is a progressive neurological disorder that has many nonmotor symptoms, particularly during the prodromal stage of the disease.[10] Some of PD's nonmotor symptoms include autonomic, neuropsychiatric, sleep, and sensory dysfunction.[11,12] Interestingly, many of these nonmotor symptoms overlap with GWI symptoms.[8] No study to date has investigated whether GW veterans are at increased risk for PD. However, there was a study funded by the Congressionally Directed Medical Research Program (CDMRP) that found the risk of PD was significantly increased among veterans who were deployed during World War II or Vietnam compared to veterans who served during peacetime. This suggests that certain military occupations and/or wartime exposures may be associated with increased risk for PD.[13] Because some GW-related exposures, particularly organophosphorus pesticides, have been linked to increased risk for PD,[14] it has been suggested that veterans with GWI may be at increased risk for PD.[8]

Sleep disturbance is a symptom that is common to both GWI,[1] prodromal PD, and PD.[15] Rapid eye movement (REM) sleep behavior disorder (RBD) is a condition where affected individuals enact their dreams because they lack the muscle paralysis (ie, atonia) that typically occurs during REM sleep.[16] Research indicates that more than 80% of patients with RBD eventually develop PD,[17] dementia with Lewy bodies (DLB),[18,19] or multiple-system atrophy (MSA).[20–23] Because PD, DLB, and MSA are all progressive neurodegenerative disorders produced by alpha-synuclein, RBD is considered a marker of prodromal alpha-synucleinopathy.[24]

The first aim of this study is to estimate the prevalence of RBD among deployed GW veterans with GWI using the 10-item REM

From the Departments of Radiology and Biomedical Imaging and Psychiatry and Behavioral Science, University of California, San Francisco, San Francisco, California (L.L.C.); San Francisco Veterans Affairs Health Care System, San Francisco, California (L.L.C., T.S., J.R.P.Z.); and Department of Neurology, University of California, San Francisco, San Francisco, California (T.S., J.R.P.Z.).
ORCID: 0000-0002-8593-2434
ORCID: 0009-0008-5824-3335
ORCID: 0000-0002-8556-5275
Funding sources: This work was supported by a VA Clinical Sciences Research and Development Award CX002522. Additional support was provided for L.L.C. by a VA Research Career Scientist Award 2IO1CX000798.
Conflict of interest: The authors alone are responsible for the views expressed in this article. They do not necessarily represent the decisions, policy, or views of their affiliated institutions. The authors declare no conflict of interest.
Specific author contributions: L.L.C. contributed to funding, study design, data preparation and analysis, and the initial and final draft of the manuscript. T.S. and J.R.P.Z. performed neurological examinations, critically reviewed all manuscript revisions, and approved the final draft of the manuscript.
Data availability: Deidentified data will be provided as requested.
EQUATER Network checklist utilized: None.
AI detailed statements: No artificial intelligence was utilized at any stage of this study. All intellectual and scientific content was developed independently by the authors.
Ethical considerations: This study was approved by the University of California, San Francisco Institutional Review Board.
Supplemental digital contents are available for this article. Direct URL citation appears in the printed text and is provided in the HTML and PDF versions of this article on the journal's Web site (www.joem.org).
Address correspondence to: Linda L. Chao, PhD, San Francisco VA Health Care System, 4150 Clement Street (114M), San Francisco, CA, 94121 (linda.chao@ucsf.edu).
Written work prepared by employees of the Federal Government as part of their official duties is, under the U.S. Copyright Act, a "work of the United States Government" for which copyright protection under Title 17 of the United States Code is not available. As such, copyright does not extend to the contributions of employees of the Federal Government.
DOI: 10.1097/JOM.0000000000003576

Behavioral Disorder Screening Questionnaire (RBDSQ),[25] a validated screening tool for RBD.[26] Because it has been suggested that GW veterans with high levels of GW deployment–related exposures may be at increased risk for PD,[8] a second aim of the study is to investigate the relationship between GW-deployment related exposures and RBDSQ scores. Animal[27,28] and human[29–32] studies have link RBD to abnormalities in the brainstem. Because neuroimaging studies of GW veterans have also reported atrophy[33–36] and other microstructural abnormalities[33–36] in the brainstems of veterans with GWI, a final aim of the study is to examine the relationship between brainstem volume, RBDSQ scores, and GW-deployment-related exposures.

MATERIALS AND METHODS

Study Design and Participants

This cross-sectional study was conducted at the San Francisco VA Health Care System (SFVAHCS) from 2020 to 2024. Participants were US veterans who were deployed to the Persian Gulf region between August 1990 and July 1991. Recruitment was conducted through direct mailings to deployed GW veterans, identified from a list requested from the Defense Manpower Data Center (DMDC), social media advertisements, and veteran referrals. It should be noted that the title of the study "An investigation of the relationship between toxicant exposures during Gulf War deployment and prodromal Parkinson's disease" may have attracted GW veterans with PD and/or PD-like symptoms. After interested veterans contacted the study team, they were provided with additional study information. If veterans were still interested in participating, they took part in a telephone screening interview to determine eligibility. Eligible veterans were invited to sign the informed consent form approved by the institutional review boards of the University of California, San Francisco (UCSF) and the SFVAHCS.

Study Outcomes

Assessment of RBD and Other Nonmotor PD Symptoms

Probable RBD was assessed with the 10-item REM Behavioral Disorder Screening Questionnaire (RBDSQ),[25] a validated screening tool for RBD that asks about the characteristics of dreams and dream-enactment behaviors.[26] Higher scores on the RBDSQ indicate a greater likelihood of RBD. RBDSQ scores ≥5 have demonstrated a sensitivity of 91% and a specificity of 77%.[26] To assess other nonmotor PD symptoms, we use the Non-Motor Symptoms Scale (NMSS),[37] a 30-item questionnaire that assesses the severity and frequency of nonmotor symptoms across nine dimensions; the Scales for Outcomes in Parkinson's Disease–Autonomic (SCOPA-AUT),[38] a self-administered questionnaire for screening autonomic symptoms in patients with PD; and the University of Pennsylvania Smell Identification Test (UPSIT),[39] a standardized, forced-choice assessment of 40 odorants. The majority of UPSIT assessments were completed remotely, following previously described procedures.[40] Research has shown that results from remotely administered UPSIT are comparable to the data collected in the clinic.[41] We also used the 15-item Geriatric Depression Scale (GDS)[42] to screen for depression, another common nonmotor symptom in PD.[43]

Classifying GWI/Chronic Multisymptom Illness Cases

Two case definitions have been recommended by the Institute of Medicine (now National Academy of Science) for use in research studies and in documenting the clinical presence of GWI.[44] These include the Centers for Disease Control and Prevention Chronic Multisymptom Illness (CDC CMI) case definition[45] and the Kansas GWI[6] case definition. Because both case definitions rely on self-reported symptoms, we used the veterans' responses to the symptom portion of the Kansas Gulf War Military History and Health Questionnaire[6] to ascertain CDC CMI[45] and Kansas GWI[6] case status. See Supplementary Table 1 (http://links.lww.com/JOM/C140) for the inclusion/exclusion criteria for each case definition.

GWI Symptom Severity

To quantify overall GWI symptom severity, we summed the veterans' responses about the severity (1 = mild, 2 = moderate, 3 = severe) of symptoms (ie, fatigue, sleep problems, somatic pain, skin abnormalities, gastrointestinal, respiratory, neurological, cognitive, and mood symptoms) used to determine Kansas GWI case status.[6] Higher GWI severity scores indicate more symptoms and/or more severe symptoms (see Supplementary Table 2, http://links.lww.com/JOM/C141)

GW Deployment–Related Exposures

Research has shown that many GW-related exposures are highly correlated.[46,47] Therefore, to avoid the potential for confounding effects of intercorrelated, concurrent exposures and to reduce the number of individual comparisons, we created a GW exposure index based the on recent work of Steele and colleagues.[48] This study examined patterns of ill health and GWI in the VA Million Veteran Program (MVP) Gulf War veteran cohort in relation to wartime exposures and key deployment features. In multivariable models, Steele and colleagues found that GWI/ill health in the MVP GW cohort (n = 14,103 veterans) was most strongly associated exposure to chemical/biological warfare agents, taking pyridostigmine bromide (PB) pills, and use of skin pesticides. Therefore, we created a GW exposure index by summing the veterans' responses on the Kansas Gulf War Military History and Health Questionnaire to questions about how frequently they heard chemical alarms sound, used PB, used cream or liquid and/or powdered pesticides, and used flea collars directly on their skin. Higher scores on the GW exposure index indicate more exposures and/or exposures of longer duration (see Supplementary Table 3, http://links.lww.com/JOM/C142)

Health History

We used a self-report questionnaire to query participants about their physical and mental health history, such as prior diagnoses of PD, traumatic brain injury (TBI), and posttraumatic stress disorder (PTSD).

Neurological Examinations

A subset of the study participants (n = 41) was examined by a board-certified neurologist (J.R.P.Z. or T.S.) as part of the study. Twenty-three neurological examines were conducted in person, prior to the pandemic lockdown. Eighteen neurological examinations were conducted remotely after the pandemic. The remote neurological examinations were conducted over Zoom following the general guidance and recommendations set forth by the American Academy of Neurology (AAN) for teleneurology visits.[49,50]

Magnetic Resonance Image Acquisition and Brainstem Volume Estimation

A subset of the participants (n = 16) was scanned at the SFVAHCS on a 3T Siemens Skyra magnetic resonance imaging (MRI) system (Erlangen, Germany) equipped with a 32-channel receiver head coil. All the participants who underwent 3T MRIs also underwent neurological examinations. A T1-weighted magnetization-prepared rapid gradient-echo (MP-RAGE) sequence was acquired with 1-mm isometric voxel size (matrix size 256 × 256 × 176), resolution of repetition time of 2500 ms, echo time of 2.98 ms, and inversion time of 1100 ms. The images were processed with cortical reconstruction, noise and distortion correction, volume segmentation, intensity normalization, and parcellations using FreeSurfer version 7.1.0 software package.[51] A FreeSurfer standard parcellation atlas (the Desikan-Killiany atlas) with anatomically labeled regions of interest (ROIs) were spatially warped from the Talairach

template onto each individual T1-weighted image. Quantitative volume measures of the brainstem, as well as cortical and subcortical brain regions and the estimated intracranial volume (eICV) were extracted from parcellated brain regions of each T1-weighted image. Supplementary Table 4 (http://links.lww.com/JOM/C143) summarizes the demographic and clinical characteristics of the different subsets of veterans with neurological examinations and 3T MRI scans.

Statistical Analysis

Demographic and military characteristics of study sample are described through means and standard deviations (SD) for continuous variables and percentages for categorical variables. Initial analyses used Mann-Whitney *U* and chi-squared tests to compare differences between veterans with and without PD diagnoses and veterans with and without probable RBD (ie, RBDSQ >5 or <5). Bivariate associations between RBDSQ, NMSS, SCOPA-AUT scores, UPSIT, GWI/CMI severity, and overall GW exposure index were examined with Spearman's rank coefficients. Finally, we used linear regression to test if RBDSQ scores, our a priori independent variable, significantly predicted brainstem volume, our a priori dependent variable. We included eICV as an independent variable to account for intersubject variations in head size. All analyses were conducted using IBM Statistical Package for the Social Sciences (SPSS) Version 29 (IBM Corporation, Armonk, NY).

Exploratory Analysis

The positive relationship between RBDSQ scores and the GW exposure index, and the group difference in the GW exposure index between veterans with and without probable RBD prompted us to explore this relationship further. We tested one model, using stepwise linear regression to determine if overall GW exposure index could predict continuous RBDSQ scores over and above age, sex, PD, PTSD, and TBI status, which were entered in the first step of the regression model. Because this analysis was exploratory, we did not adjust for multiple comparisons. However, subsequent studies with preplanned hypotheses will be needed to confirm any observed associations.[52]

RESULTS

Study Population

Table 1 summarizes the demographic, military, and health characteristics of the entire study sample and stratified by PD status. Fifty veterans in the sample had PD. Forty-eight veterans self-reported

TABLE 1. Demographic, Military, and Health Characteristics of Study Sample

	All	No PD	PD	RBDSQ < 5	RBDSQ ≥ 5
N	165	115	50	41	124
Age[a] (yr), mean (SD)	61.6 (7.3)	60.7 (6.7)	63.8 (8.0)	63.1 (8.2)	61.1 (6.9)
Sex, *n* (%)					
Female	10 (6)	5 (4)	5 (10)	5 (12)	5 (4)
Male	155 (94)	110 (96)	45 (90)	36 (88)	119 (96)
Race, *n* (%)					
Black/African American	6 (4)	5 (4)	1 (2)	2 (5)	4 (3)
Caucasian/White	142 (86)	101 (88)	41 (82)	36 (88)	106 (86)
Other and/or >1 race	3 (2)	2 (2)	1 (2)	0 (0)	3 (2)
Not reported/declined to state	14 (8)	7 (6)	7 (14)	3 (7)	11 (9)
Ethnicity, *n* (%)					
Hispanic	15 (9)	11 (10)	4 (8)	5 (12)	10 (8)
Not Hispanic	131 (79)	97 (84)	33 (66)	33 (81)	97 (78)
Not reported/declined to state	19 (12)	7 (6)	13 (26)	3 (7)	17 (14)
Education (yr), mean (SD)	14.6 (2.1)	14.3 (2.0)	15.3 (2.2)	14.5 (2.2)	14.6 (2.1)
Rank,[b] *n* (%)					
Enlisted	138 (83)	101 (88)	37 (74)	34 (83)	104 (84)
Officer	26 (16)	13 (11)	13 (26)	6 (15)	20 (16)
Not reported	1 (1)	1 (1)		1 (2)	
Branch of service,[b] *n* (%)					
Army	102 (62)	69 (60)	33 (66)	25 (61)	77 (62)
Navy	22 (13)	13 (11)	9 (18)	5 (12)	17 (14)
Air Force	16 (10)	12 (11)	4 (8)	4 (10)	12 (10)
Marines	25 (15)	21 (18)	4 (8)	7 (17)	18 (14)
Component of service,[b] *n* (%)					
Active Duty	125 (76)	89 (77)	36 (72)	27 (66)	98 (79)
Reserves	24 (14)	18 (16)	6 (12)	9 (22)	15 (12)
National Guard	14 (9)	7 (6)	7 (14)	4 (10)	10 (8)
Not reported	2 (1)	1 (1)	1 (2)	1 (2)	1 (1)
CDC CMI case (%)	100	100	100	100	100
Met Kansas GWI case definition, *n* (%)	69 (42)	69 (60)	—	22 (54)	47 (38)
Had Kansas GWI exclusionary condition(s), *n* (%)	96 (58)	41 (38)	50 (100)	19 (46)	77 (62)
GWI severity index	40.4 (15.7)	40.7 (15.7)	39.8 (15/8)	29.8 (12.1)	44.0 (15.2)
GW exposure index[c]	6.1 (3.8)	6.3 (3.8)	5.6 (3.7)	4.4 (2.7)	6.7 (3.9)
Self-reported PTSD, *n* (%)	27 (16)	22 (19)	6 (5)	1 (2)	26 (21)
Self-reported TBI, *n* (%)	7 (4)	5 (10)	1 (2)	0 (0)	7 (6)

CDC CMI, Centers for Disease Control and Prevention Chronic Multisymptom Illness; GWI, Gulf War Illness; PTSD, Posttraumatic stress disorder; TBI, Traumatic brain injury.

[a]At the time of study.

[b]During the Gulf War.

[c]Frequency of using pyridostigmine bromide (PB) cream/liquid, or powdered pesticides on skin, flea collars, and hearing chemical alarms sound during GW deployment.

previous PD diagnosis. Study neurologists confirmed 17 of these veterans' self-reports of PD during the neurological examinations. Two veterans who did not self-report PD were diagnosed with probable PD during the neurological examinations.

Compared to veterans without PD, those with PD were older (U = 2152, P = 0.01) and had more years of formal education (U = 2087.5, P = 0.006). There were no differences in sex (χ^2 = 1.96, df = 1, P = 0.16) or race (χ^2 = 3.25, df = 3, P = 0.36). However, there was a group difference in ethnicity (χ^2 = 12.98, df = 2, P = 0.002). More veterans with PD had missing ethnicity data compared to veterans without PD (26% vs 6%). There was a marginal difference in rank (χ^2 = 6.01, df = 2, P = 0.05). There were slightly more officers among veterans with PD. There were no group differences in other military characteristics (service branch: χ^2 = 4.01, df = 3, P = 0.26; service component: χ^2 = 3.39, df = 3, P = 0.34). There were no differences in the rates of self-reported PTSD (χ^2 = 2.12, df = 1, P = 0.15) or TBI (χ^2 = 0.89, df = 1, P = 0.35) between veterans with and without PD. All veterans in the study met the criteria for CDC CMI. There were no differences in GWI severity index (U = 2710, P = 0.56) or GW exposure index (U = 2599, P = 0.33) between veterans with and without PD. Supplementary Table 5 (http://links.lww.com/JOM/C144) summarizes GWI severity, GW exposure indices, and measures of nonmotor symptoms as a function of PD, Kansas GWI case, and possible RBD status.

As noted previously, a subset of the GW veterans (n = 41) underwent neurological examinations as part of study procedures. The presence of PD was confirmed in 17 veterans who self-reported PD, and absence of PD was confirmed in 16 veterans who did not self-report PD. Study neurologists made eight diagnoses in veterans who did not self-report any medical conditions: Two veterans were diagnosed with probable PD, one veteran with vascular PD, one with psychogenic PD, two with essential tremor, one with adult tics syndrome, and one veteran with possible RBD. Interestingly, the veteran who was diagnosed with possible RBD had the maximum possible score on the RBDSQ, to which the study neurologist was blind.

The first aim of this study was to estimate the prevalence of RBD among veterans with GWI using an RBDSQ cutoff score of ≥5. Table 1 also summarizes the characteristics of the study sample stratified by probable RBD status (ie, RBDSQ ≥5). Three quarters (75%) of the sample had probable RBD. There was no difference in the percentage of probable RBD cases among veterans with (80%) and without (73%) PD (χ^2 = 0.90, df = 1, P = 0.34). There were also no group differences in age (U = 2183, P = 0.18), education (U = 2418.5, P = 0.81), race (χ^2 = 1.33, df = 3, P = 0.72), ethnicity (χ^2 = 1.64, df = 2, P = 0.44), or military characteristics (rank: χ^2 = 3.07, df = 2, P = 0.22; service branch: χ^2 = 0.19, df = 3, P = 0.98; service component: χ^2 = 3.55, df = 3, P = 0.32) between veterans with and without probable RBD. There were marginally more males (χ^2 = 3.61, df = 1, P = 0.06) and marginally fewer Kansas GWI cases (χ^2 = 3.14, df = 1, P = 0.08) among veterans with probable RBD. There was a significant group difference in GWI severity index (U = 1233.5, P < 0.001, Cohen's d = 0.9). Veterans with probable RBD also had higher GW exposure index (U = 1657, P < 0.001, Cohen's d = 0.6) compared to veterans without probable RBD (see Fig. 1). More veterans with probable RBD self-reported PTSD (χ^2 = 7.73, df = 1, P = 0.005). Although all veterans who self-reported TBI had probable RBD, the group difference for TBI did not reach significant (χ^2 = 2.42, df = 1, P = 0.12).

Because estimating the likelihood of RBD solely on a RBDSQ cutoff score of 5 or greater without overnight polysomnography may artificially inflate the number of positive cases, we also examined the data using stricter RBDSQ thresholds (ie, >6 and >7). Supplementary Table 6 (http://links.lww.com/JOM/C145) and Supplementary Table 7 (http://links.lww.com/JOM/C146) show that, even with the stricter RBDSQ thresholds, similar directions and magnitudes persisted in the effect that probable RBDSQ had on GWI severity and GW exposures. Because 73% of the veterans without PD had RBDSQ scores ≥5, 65% had RBDSQ scores ≥6, and 60% had RBDSQ scores ≥7, this supports our hypothesis that veterans with GWI are at increased risk for prodromal PD and PD, possibly through RBD as a mediator.

The second aim of the study was to examine the relationship between RBDSQ scores and GW-related exposures, quantified with the GW exposure index. The GW exposure index was positively correlated with RBDSQ scores in the entire sample and in various subgroups of veterans (eg, those with and without PD, those with and without Kansas GWI exclusionary conditions). See Figure 2 and Table 2. The GW exposure index also correlated positively with NMSS and SCOPA-AUT scores; however, it was not correlated with UPSIT scores.

The positive relationship between RBDSQ scores and GW exposure index, and the significant difference in GW exposure index between veterans with and without probable RBD (based on thresholds of ≥5, ≥6, and >7) prompted us to examine this relationship further in exploratory analyses. With the intention of generating hypotheses for future research studies, we tested one exploratory model (summarized in Table 3). Even after accounting for demographic (age, male sex) and clinical (PD, PTSD, and TBI status) variables in step 1 of the model, the GW exposure index significantly predicted RBDSQ scores in step 2 of the model (standardized β = 0.317, t = 4.29, P < 0.001). We did not adjust for multiple comparisons because the analysis was exploratory. However, future studies with preplanned hypotheses will be needed to confirm the observed associations between GW exposure levels and RBDSQ scores.[52]

Although only 16 veterans had MRI data, the final aim of this study was to investigate the relationship between brainstem volume (the a priori dependent variable) and RBDSQ scores (the a priori independent variable). We included eICV as an independent variable to

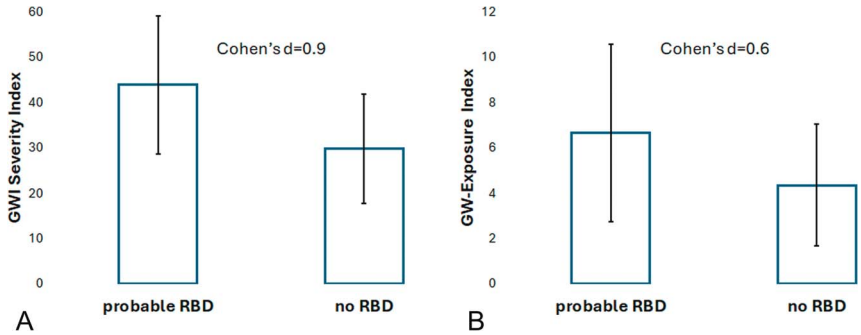

FIGURE 1. Differences in (A) GWI severity index and (B) GW exposure index between veterans with and without probable REM behavioral sleep disorder (RBD). Standardized effect sizes (Cohen's d) are also reported. Error bars are standard deviations.

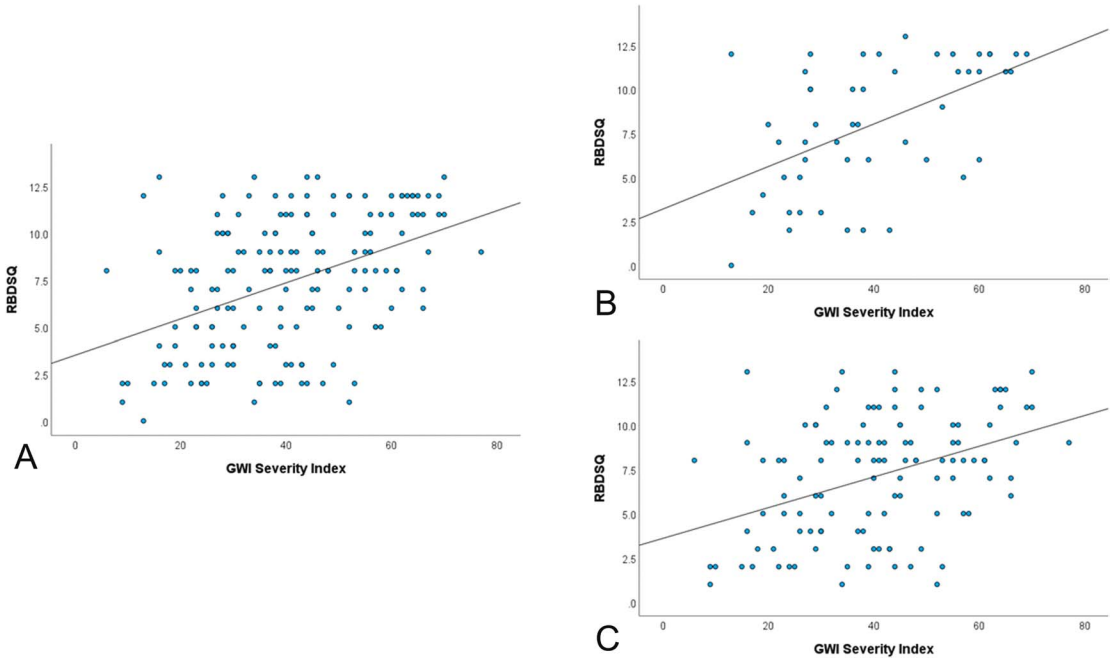

FIGURE 2. Relationship between GWI severity index and the RBDSQ in (A) the entire study sample and in the subset of veterans (B) with PD and (C) without PD diagnoses.

account for intersubject variations in head size. Descriptive statistics of the regression are presented in Table 4. The regression was significant ($F_{2,13} = 9.03$, $P = 0.003$) with an R^2 of 0.581, indicating that RBDSQ and eICV together explained approximately 58.1% of the variance in brainstem volume. The regression equation was as follows: brainstem volume = 11,953.03 + (−381.93) × (RBDSQ) + 0.01 × eICV. This suggests that for each 1-point increase in RBDSQ, predicted brainstem volume decreases by approximately 381.98 mm^3. Confidence intervals indicated that we can be 95% certain that the slope to predict brainstem volume from RBDSQ is between −711.83 and −52.14.

DISCUSSION

It has previously been suggested that veterans with GWI may be at increased risk for PD because some of the chemicals present in the GW milieu (eg, organophosphorus pesticides, chemical warfare agents like sarin, organochlorines, permethrin, and PB) have been linked to increased risk for both GWI[1,4–7] and PD.[8] Notably cholinergic dysfunction,[53,54] oxidative stress,[55,56] and neuroinflammation[57,58] have all been cited as biological mechanisms involved both diseases. It is notable that many GWI symptoms overlap with nonmotor PD symptoms (eg, sleep disturbance, autonomic dysfunction, cognitive and mood impairments).[8] Because RBD is an established marker of prodromal PD,[17] we used the RBDSQ to examine the prevalence of possible RBD in deployed GW veterans with GWI. To the best of our knowledge, this is the first study to investigate rates of probable RBD in GW veterans.

We did not find any difference in the percentage of probable RBD cases between veterans with (80%) and without PD (73%). This was the case even when we explored the data using stricter RBDSQ thresholds (ie, RBDSQ ≥6 and >7). Among veterans without PD, there was no difference in the rate of probable RBD cases between veterans who met (68%) and did not meet (80%) the Kansas GWI case definition.

In the general population, the prevalence of RBD has been estimated to be between 0.74% and 2.01%.[59,60] Veterans tend to have higher rates of RBD than the general population: One study estimated a 9% prevalence of RBD among veterans seeking treatment at a sleep

TABLE 2. Spearman's Correlations Between GW Exposure Index and Measures of Nonmotor PD Symptoms

			No PD		
	All[a] (n = 165)	PD (n = 50)	All Without PD (n = 115)	GWI With Kansas Exclusionary Condition(s) (n = 46)	GWI Who Met Kansas Case Definition (n = 69)
RBDSQ	**0.331***	**0.408****	**0.318***	**0.324*****	**0.307****
NMSS	**0.268***	**0.381****	**0.226*****	0.238	0.234******
SCOPA-Aut	**0.241****	0.155	**0.263****	0.266******	**0.252*****
UPSIT[b]	0.069	0.105	0.040	−0.096	0.159
GDS[c]	**0.172*****	0.190	0.140	0.162	0.144

GDS, Geriatric Depression Scale; NMSS, Non-Motor Symptoms Scale; RBDSQ, REM Sleep Behavioral Disorder Screening Questionnaire; SCOPA-Aut, Scales for Outcomes in Parkinson's Disease–Autonomic Dysfunction; UPSIT, University of Pennsylvania Smell Identification Test.

[a]All met the criteria for CDC CMI case definition.

[b]Data available for 120/165 subjects: 29 PD, 91 no PD; 54 KS GWI cases.

[c]Data available for 159/165 subjects: 49 PD, 110 no PD; 69 KS GWI cases.

Significant values are bolded: *$p < 0.001$, **$p \leq 0.01$, ***$p < 0.05$, ****$p < 0.08$.

TABLE 3. Summary of Exploratory Hierarchical Regression Analysis of RBDSQ Scores

Independent Variables	ΔR^2	Adjusted R^2	ΔF	Standardized Coefficients β	
Step 1	0.103	0.075	3.67**	Step 1	Step 2
Age				−0.085	−0.067
Male				0.093	0.122
PTSD				0.147****	0.108
TBI				0.215**	0.154***
PD				0.175***	0.191**
Step 2	0.197	0.166	18.41*		
GW exposure index					0.317*

GW, Gulf War; PD, Parkinson's disease; PTSD, Posttraumatic stress disorder; TBI, Traumatic brain injury.
*$p < 0.001$, **$p \leq 0.01$, ***$p < 0.05$, ****$p < 0.06$.

center.[61] The 68% to 80% of GW veterans in this study who do not have PD but who had RBDSQ scores ≥5 are much higher than the rate of RBD among treatment-seeking veterans and in the general population. However, it should be noted that we did not validate the presence of RBD with overnight polysomnography, and RBDSQ may not be able to differentiate RBD from trauma-associated sleep disorder (TASD), a parasomnia associated with PTSD.[62] For this reason, RBDSQ may have limitations in the assessment of veterans, particularly among those with high rates of PTSD and other trauma-related conditions. With these caveats in mind, our finding that the majority of GW veterans without PD had high RBDSQ scores (73% had RBDSQ ≥5, 65% had RBDSQ ≥6, 60% had RBDSQ ≥7) supports our hypothesis that veterans with GWI are at increased risk for prodromal PD and PD, possibly through RBD as an intermediary. It is noteworthy that the study neurologist, who was blinded to the veteran's RBDSQ, diagnosed one veteran with possible RBD during the neurological examination.

Consistent with the literature on sex differences in RBD,[63] there were marginally more male than female GW veterans with probable RBD. In line with reports of a high incidence of comorbidity of RBD with PTSD[61,64] and TBI,[61,65] more veterans with probable RBD self-reported PTSD compared to veterans without probable RBD. Furthermore, all seven veterans who self-reported TBIs had probable RBD.

It has been suggested that GW veterans with higher levels of GW deployment–related exposures may be at increased risk for PD.[8] To the extent that high RBDSQ scores is a sign of prodromal PD,[17] we found that the GW exposure index was positively and significantly correlated with RBDSQ scores. In exploratory analyses, GW exposure index significantly predicted RBDSQ scores over and above age, sex, PD, PTSD, and TBI status. We also found positive correlations between the GW exposure index and other measures of nonmotor PD symptoms (NMSS and SCOPA-AUT) in the entire study sample and in subgroups of veterans with and without PD and in subgroups of veterans who met the Kansas GWI case definition.

Other studies of GW veterans have reported a positive relationship between the amount/extent of GW-related exposures and GWI symptom severity. For example, Sullivan et al[66] found that GW veterans with the highest combined exposure to pesticides and PB had the worst cognitive performance. More recently, Krengel et al[67] reported that GW pesticide applicators with higher levels of self-reported GW exposures had more health symptom complaints and medical conditions than GW military personnel with low pesticide exposure. The current finding adds to the literature by providing suggestive evidence that greater GW exposures were associated with higher RBDSQ scores, which, in turn, is considered an early marker of prodromal PD.

The third major finding of the study is that brainstem volume was inversely related to RBDSQ scores. One caveat is that we only had neuroimaging data (ie, brainstem volume) for 16 veterans in the study. Nevertheless, animal,[27,28] human lesion,[29–32] and human imaging studies[68,69] have linked RBD to structural and connectivity abnormalities in the brainstem. The main brainstem structures involved in the paralysis of postural muscles during REM sleep (ie, REM atonia) are the sublaterodorsal nucleus in the pons and the ventromedia medulla. These structures normally work together to inhibit motor neurons, preventing movement during REM sleep.[70] In RBD, damage to or degeneration in the dorsal pons and ventromedial medulla leads to a loss of REM atonia, which allows individuals to physically act out their dreams.[71] In this respect, it is notable that the brainstem is one of the earliest structures affected by PD pathology.[72] Diffusion tensor imaging (DTI) studies have reported microstructural abnormalities in the brainstems of patients with PD and RBD.[73,74] Interestingly, brainstem abnormalities have also been noted in veterans with GWI.[33–36] Recently, Zhang et al[36] reported significant correlations between poor sleep quality, brainstem volume and measures of axonal integrity and myelination in the brainstem of veterans with GWI. Autonomic dysfunction is common in veterans with GWI[53,75–77] and patients with RBD.[78] Interestingly, autonomic dysfunction is also a major category of nonmotor PD symptoms.[79] Because the brainstem is the control center for the autonomic nervous system,[80] brainstem-related autonomic dysfunction may be the common factor linking GWI, RBD, and PD. In support of this, we found a positive and significant correlation between the SCOPA-AUT, a questionnaire specifically designed to screen for autonomic symptoms in patients with PD,[38] and the GW exposure index.

Multiple case definitions have been used to define GWI in the decades since the end of the Gulf War because there is no consensus

TABLE 4. Summary of Linear Regression Analysis on Brainstem Volume

Independent Variables	R^2	Adjusted R^2	ΔF	Unstandardized Coefficients	Standardized Coefficients β	95% Confidence Interval	
eICV	0.380	0.336	8.58*	0.009	0.479	0.001	0.017
RBDSQ	0.581	0.517	6.26**	−381.98	−0.469	−711.83	−52.14

eICV, estimated intracranial volume; RBDSQ, REM Behavioral Disorder Screening Questionnaire score.
*$p = 0.01$, **$p = 0.03$.

GWI case definition.[44,81] The Institute of Medicine recommended use of CDC CMI and the Kansas GWI case definitions for clinical and research purposes, respectively.[44] However, Haley and colleagues[82] have published reports using symptom complexes derived from factor analysis of symptom data to define/categorize GWI. Haley Syndrome 1 (impaired cognition) is characterized by attention, memory, and sleep problems along with depression. Haley Syndrome 2 (confusion/ataxia) involves cognitive and balance problems. Haley Syndrome 3 (neuropathic pain) involves joint and muscle pain.[1,82] It is notable that Haley et al[53] have suggested that GWI symptoms in all THREE syndrome groups, but especially in Syndrome 2, are due to cholinergic autonomic dysfunction resulting from sarin exposure. RBD has been linked to the degeneration of brainstem structures, particularly in the mesopontine cholinergic system that is involved in regulating muscle tone during REM sleep.[83] Unfortunately, we did not collect sufficient information to use factor analysis to categorize veterans in the current study into the Haley syndromes. Future research will be needed to investigate the prevalence of RBD in GW veterans with Haley syndromes 1, 2, and 3.

Although the Institute of Medicine recommended use of the CDC CMI and Kansas GWI case definitions for clinical and research purposes,[44] both definitions were developed in the late 1990s to capture profile of GWI in the years shortly after the end of the Gulf War. Results of recent studies have raised concerns that these case definitions may no longer adequately reflect the current profile of veterans with GWI.[84–86] For example, the Kansas GWI exclusionary conditions were initially meant to exclude veterans with medical and psychiatric diagnoses that did not fall under the general rubric of GWI but might predict similar symptoms as GWI and/or might interfere with the veterans' perceptions of or ability to report their symptoms. However, recent studies have found that the number and severity of GW veterans' symptoms and comorbid diagnoses have continued to increase as GW veterans age.[87–89] For this reason, more veterans with GWI are developing conditions that are considered exclusionary for the Kansas GWI case definition.

We chose to enroll GW veterans with PD, even though PD is a Kansas GWI exclusionary condition, because we wanted to examine potential demographic and deployment differences between veterans with and without PD. Because we included veterans with one Kansas GWI exclusionary condition, it was difficult to justify omitting veterans with other Kansas GWI exclusionary conditions. Moreover, we wanted to enroll veterans with a wide range of exposures to all possible GW-related chemicals. Because some GW-related chemicals have been linked to increased risk for cancer,[90,91] we did not want to inadvertently restrict the range of GW-related exposures in our study sample by excluding GW veterans who had cancer. For these reasons, we included all deployed GW veterans who were interested in participating in the study and who were able to provide informed consent.

Besides PD, the other Kansas GWI exclusionary conditions that veterans had in the current study had included cancer and melanoma in the past 3 years, type 2 diabetes with poor glycemic control in the past 2 years, heart disease in the past 2 years, minor stroke, rheumatoid arthritis, liver and kidney disease in the past 3 years, hospitalization for PTSD, depression, and/or alcohol or substance abuse in the past 2 years. Not surprisingly, Kansas GWI cases (ie, veterans without these comorbidities) had lower GWI severity indices than veterans with Kansas GWI exclusionary condition(s). Despite these comorbidities, there were no significant differences in NMSS, SCOPA-AUT, UPSIT, or GDS scores between veterans without PD who did and did not have Kansas GWI exclusionary condition(s). There were also no significant differences in the GW exposure index and the percentage of veterans with possible RBD, and the positive relationship between the GW exposure index and RBDSQ scores was present in all subgroups, regardless of PD or presence/absence of Kansas GWI exclusionary condition(s). This suggests that the relationship between GW-related exposures and risk for probable RBD (and prodromal PD) is not a by-product of comorbid Kansas GWI exclusionary conditions, and that these findings can be generalized to "real-world" GW veterans with GWI.

Age is the most significant risk factor for PD,[92] and high levels of education have been reported to be a PD risk factor.[93,94] Therefore, it is not surprising that age and education differed significantly between veterans with and without PD. There was also a group difference in ethnicity because there were more missing data for this demographic variable in the PD than non-PD group. Some multiethnic studies have reported a higher incidence of PD in Latinos compared to White non-Latino (WNL),[95,96] whereas other studies have reported similar or even lower prevalence of PD in Latinos compared to the general US population.[97,98] A recent large community-based study of PD in rural California reported that Latino patients had higher frequencies of early-onset PD and worse motor and nonmotor symptoms compared to WNL.[99] Future research will be necessary to determine if ethnicity plays a role in risk for PD among deployed GW veterans.

One explanation for the lack of difference in other variables between veterans with and without PD is that some of the veterans who did not self-report PD diagnoses have prodromal PD, as evidenced by their high scores on the RBDSQ. In fact, two veterans who did not self-report PD were diagnosed with probable PD by study neurologists during the neurological examination. Unfortunately, we were unable to administer neurological examinations to all veterans enrolled in the study. Future longitudinal follow-up will be necessary to determine if GW veterans with high RBDSQ scores in the study sample develop PD.

Besides RBD, a reduced sense of smell (ie, hyposmia) is another well-documented nonmotor symptom of prodromal PD.[100] A previous study found that GW veterans with GWI have hyposmia.[40] Although there is suggestive evidence that RBD is associated with impaired olfactory function,[101,102] we did not find significant differences in UPSIT scores between veterans with and without possible RBD in the current study. There was also no significant correlation between UPSIT and RBDSQ scores. Even though hyposmia is nonmotor symptom of prodromal PD,[100] it is not specific to prodromal PD or PD. One in three elderly individuals has olfactory loss due to other etiologies, and many pathologies other than PD can result in forms of hyposmia.[103] The olfactory neuroepithelium is particularly vulnerable to environmental exposures.[104,105] Because GW veterans experienced so many environmental exposures during their 1990–1991 deployment to the Persian Gulf region, even if those environmental exposures did not increase the veterans' vulnerability for developing PD, it is possible that it may have contributed to the veterans' hyposmia.

This study had some limitations that warrant consideration: First, the cohort consisted entirely of GW veterans who signed up for in a research study examining the relationship between toxicant exposures during Gulf War deployment and prodromal Parkinson's disease. This may have skewed recruitment toward GW veterans with PD and/or more PD-like symptoms. Second, there were more participants with PD in the subsets of veterans who had neuroexaminations and MRIs. This could be because veterans with PD are less likely to be working full time and therefore had greater availability to come to the San Francisco VA for the in-person study procedures before the pandemic lockdown compared to veterans without PD. It may also be possible that the veterans with PD were more motivated to participate in all study procedures compared to veterans without PD, who mostly opted to participate in remote study procedures only. Third, even though the Institute of Medicine recommended the use of the Kansas GWI case definition for research purposes,[44] we enrolled study participants who had Kansas GWI exclusionary conditions. This is because research suggests that the number and severity of GW veterans' symptoms and comorbid diagnoses have continued to increase as GW veterans age.[87–89] Thus, fewer and fewer GW veterans are eligible for GWI research studies that use the strict Kansas GWI case definition. Because we enrolled "all comers," it could be argued that the

current study sample is more representative of "real-world" GW veterans with GWI than the GW veterans who participate in research studies that utilize the restrictive Kansas GWI case definition. Another limitation of this, and all existing research on toxicant exposures in GW veterans, is that we based our estimates of GW deployment–related exposures on self-report data, collected decades after the GW, which could have introduced recall bias. However, no data were collected about which GW personnel was exposed to which potentially hazardous substances and at what levels during the Gulf War. Therefore, self-report is one of the only means to study the effects of GW-related toxicant exposures. Furthermore, there are no objective biomarkers of GW deployment–related toxicants that can assess exposure 30 years after the fact. A fifth limitation of the study is that risk for RBD was determined solely by a cutoff score on the RBDSQ. We did not verify RBD with overnight polysomnography or collect detailed sleep history from study participants. Also, as noted earlier, the RBDSQ may not be able to differentiate RBD from TASD. Therefore, this may have confounded the findings of a high percentage of veterans with GWI having RBD based solely on an RBDSQ cutoff score. A sixth limitation is that only 41/165 veterans were examined by study neurologists. The remaining veterans' self-reports of PD, PTSD, and TBI were not verified by clinical diagnostic interviews. A final limitation is that only small subset of veterans underwent 3T neuroimaging. Thus, our finding that brainstem volume was inversely related to RBDSQ scores may not generalize beyond the 16 veterans with imaging data, the majority of whom had PD. Despite this limitation, we can use this finding to generate hypotheses that can be tested in future research. These limitations, notwithstanding results from this study, suggest that deployed GW veterans with GWI have a high likelihood of RBD and therefore of prodromal PD. Risk of prodromal PD is increased in GW veterans who self-reported higher levels of GW-related deployment exposures.

CONCLUSION

This cross-sectional study presented evidence of a higher-than-expected rate of probable RBD in a self-selected sample of deployed GW veterans with GWI, some of whom also met the criteria for the Kansas GWI case definition. Because RBD is considered a marker of prodromal PD,[17] it may be prudent for the VA to screen deployed GW veterans with the RBDSQ and to clinically monitor (with overnight sleep studies) or refer to neurologists GW veterans whose RBDSQ score is ≥5. Several studies have shown associations between modifiable lifestyle factors and PD risks and outcomes.[106] For example, physical activity,[107,108] coffee and caffeine intake,[109] and the Mediterranean diet[110] have been associated with a reduced risk of developing PD. There is also suggestive evidence that naturally derived antioxidant phytochemicals may be beneficial for patients with PD and/or prodromal PD through their neuroprotective mechanisms targeting oxidative stress, neuroinflammation, abnormal protein accumulation, mitochondrial dysfunction, endoplasmic reticulum stress, neurotrophic factor deficit, and apoptosis.[111] Thus, GW veterans with high RBDSQ scores could be counseled to adopt lifestyle changes that may potentially lower their risk for developing PD.[112] The VA can also offer these at-risk GW veterans neuroprotective therapies when they become available.

ACKNOWLEDGMENTS

The authors thank the Gulf War veterans who took the time to participate in this research.

REFERENCES

1. White RF, Steele L, O'Callaghan JP, et al. Recent research on Gulf War illness and other health problems in veterans of the 1991 Gulf War: effects of toxicant exposures during deployment. *Cortex* 2016;74:449–475.
2. Research Advisory Committee on Gulf War Veterans' Illnesses. Gulf War Illness and the Health of Gulf War Veterans: Research Update and Recommendations, 2009–2013. U.S. Government Printing Office Washington, D.C. 2014. https://www.va.gov/RAC-GWVI/RACReport2014Final.pdf. Accessed October 20, 2025.
3. National Academies of Sciences, Engineering, and Medicine. Gulf War and Health: Volume 10: Update of Health Effects of Serving in the Gulf War, 2016. Washington, DC: The National Academies Press; 2016.
4. Unwin C, Blatchley N, Coker W, et al. Health of UK servicemen who served in Persian Gulf War. *Lancet* 1999;353:169–178.
5. Kang HK, Mahan CM, Lee KY, Magee CA, Murphy FM. Illnesses among United States veterans of the Gulf War: a population-based survey of 30,000 veterans. *J Occup Environ Med* 2000;42:491–501.
6. Steele L. Prevalence and patterns of Gulf War illness in Kansas veterans: association of symptoms with characteristics of person, place, and time of military service. *Am J Epidemiol* 2000;152:992–1002.
7. Steele L, Furlong CE, Richter RJ, et al. PON1 status in relation to Gulf War illness: evidence of gene-exposure interactions from a multisite case-control study of 1990–1991 Gulf War weterans. *Int J Environ Res Public Health* 2024;21:964.
8. Chao LL. Do Gulf War veterans with high levels of deployment-related exposures display symptoms suggestive of Parkinson's disease? *Int J Occup Med Environ Health* 2019;32:503–526.
9. Willis AW, Roberts E, Beck JC, et al. Incidence of Parkinson disease in North America. *NPJ Parkinsons Dis* 2022;8:170.
10. Postuma RB, Aarsland D, Barone P, et al. Identifying prodromal Parkinson's disease: pre-motor disorders in Parkinson's disease. *Mov Disord* 2012;27:617–626.
11. Postuma RB, Berg D. Advances in markers of prodromal Parkinson disease. *Nat Rev Neurol* 2016;12:622–634.
12. Heinzel S, Berg D, Gasser T, et al. Update of the MDS research criteria for prodromal Parkinson's disease. *Mov Disord* 2019;34:1464–1470.
13. Nelson LM. Military Service and Parkinson's Disease. 2013. Available at: https://apps.dtic.mil/sti/tr/pdf/ADA583477.pdf. Accessed September 26, 2024.
14. Wang A, Cockburn M, Ly TT, Bronstein J, Ritz B. The association between ambient exposure to organophosphates and Parkinson's disease risk. *Occup Environ Med* 2015;71:275–281.
15. Bohnen NI, Hu MTM. Sleep disturbance as potential risk and progression factor for Parkinson's disease. *J Parkinsons Dis* 2019;2019:603–614.
16. Olson EJ, Boeve BF, Silber MH. Rapid eye movement sleep behaviour disorder: demographic, clinical and laboratory findings in 93 cases. *Brain* 2000;123(Pt 2):331–339.
17. Postuma RB. Prodromal Parkinson's disease—using REM sleep behavior disorder as a window. *Parkinsonism Relat Disord* 2014;20(Suppl 1):S1–S4.
18. Berg D, Postuma RB, Bloem B, et al. Time to redefine PD? Introductory statement of the MDS Task Force on the definition of Parkinson's disease. *Mov Disord* 2014;29:454–462.
19. Postuma RB, Gagnon JF, Vendette M, Montplaisir JY. Idiopathic REM sleep behavior disorder in the transition to degenerative disease. *Mov Disord* 2009;24:2225–2232.
20. Postuma RB, Iranzo A, Hogl B, et al. Risk factors for neurodegeneration in idiopathic rapid eye movement sleep behavior disorder: a multicenter study. *Ann Neurol* 2015;77:830–839.
21. Schenck CH, Boeve BF, Mahowald MW. Delayed emergence of a parkinsonian disorder or dementia in 81% of older males initially diagnosed with idiopathic REM sleep behavior disorder (RBD): 16 year update on a previously reported series. *Sleep Med* 2013;14:795–806.
22. Iranzo A, Fernandez-Arcos A, Tolosa E, et al. Neurodegenerative disorder risk in idiopathic REM sleep behavior disorder: study in 174 patients. *PloS One* 2014;9:e89741.
23. Postuma RB, Gagnon JF, Vendette M, Fantini ML, Massicotte-Marquez J, Montplaisir J. Quantifying the risk of neurodegenerative disease in idiopathic REM sleep behavior disorder. *Neurology* 2009;72:1296–1300.
24. Miglis MG, Adler CH, Antelmi E, et al. Biomarkers of conversion to α-synucleinopathy in isolated rapid-eye-movement sleep behaviour disorder. *Lancet Neurol* 2021;20:671–684.
25. Stiasny-Kolster K, Mayer G, Schafer S, Moller JC, Heinzel-Gutenbrunner M, Oertel WH. The REM sleep behavior disorder screening questionnaire—a new diagnostic instrument. *Mov Disord* 2007;22:2386–2393.
26. Li K, Li SH, Su W, Chen HB. Diagnostic accuracy of REM sleep behaviour disorder screening questionnaire: a meta-analysis. *Neurol Sci* 2017;38:1039–1046.
27. Boeve BF, Silber MH, Saper CB, et al. Pathophysiology of REM sleep behaviour disorder and relevance to neurodegenerative disease. *Brain* 2007;130:2770–2788.
28. Iranzo A. The REM sleep circuit and how its impairment leads to REM sleep behavior disorder. *Cell Tissue Res* 2018;373:245–266.
29. Tang WK, Hermann DM, Chen YK, et al. Brainstem infarcts predict REM sleep behavior disorder in acute ischemic stroke. *BMC Neurol* 2014;14:88.

30. Culebras A, Moore JT. Magnetic resonance findings in REM sleep behavior disorder. *Neurology* 1989;39:1519–1523.
31. Kimura K, Tachibana N, Kohyama J, Otsuka Y, Fukazawa S, Waki R. A discrete pontine ischemic lesion could cause REM sleep behavior disorder. *Neurology* 2000;55:894–895.
32. Xi Z, Luning W. REM sleep behavior disorder in a patient with pontine stroke. *Sleep Med* 2009;10:143–146.
33. Rayhan RU, Stevens BW, Raksit MP, et al. Exercise challenge in Gulf War illness reveals two subgroups with altered brain structure and function. *PloS One* 2013;8:e63903.
34. Christova P, James LM, Engdahl BE, Lewis SM, Carpenter AF, Georgopoulos AP. Subcortical brain atrophy in Gulf War illness. *Exp Brain Res* 2017;235: 2777–2786.
35. Zhang Y, Avery T, Vakhtin AA, et al. Brainstem atrophy in Gulf War illness. *Neurotoxicology* 2020;78:71–79.
36. Zhang Y, Vakhtin AA, Dietch J, et al. Brainstem damage is associated with poorer sleep quality and increased pain in Gulf War illness veterans. *Life Sci* 2021;280:119724.
37. Chaudhuri KR, Martinez-Martin P, Schapira AH, et al. International multicenter pilot study of the first comprehensive self-completed nonmotor symptoms questionnaire for Parkinson's disease: the NMSQuest study. *Mov Disord* 2006;21:916–923.
38. Visser M, Marinus J, Stiggelbout AM, Van Hilten JJ. Assessment of autonomic dysfunction in Parkinson's disease: the SCOPA-AUT. *Mov Disord* 2004;19: 1306–1312.
39. Doty RL, Shaman P, Dann M. Development of the University of Pennsylvania Smell Identification Test: a standardized microencapsulated test of olfactory function. *Physiol Behav* 1984;32:489–502.
40. Chao LL. Olfactory and cognitive decrements in 1991 Gulf War veterans with Gulf War illness/chronic multisymptom illness. *Environ Health* 2024;23:14.
41. Jennings D, Siderowf A, Stern M, et al. Imaging prodromal Parkinson disease: the Parkinson associated risk syndrome study. *Neurology* 2014;83:1739–1746.
42. Yesavage JA, Brink TL, Rose TL, et al. Development and validation of a geriatric depression screening scale: a preliminary report. *J Psychiatr Res* 1982;17:37–49.
43. Marsh L. Depression and Parkinson's disease: current knowledge. *Curr Neurol Neurosci Rep* 2013;13:409.
44. Committee on the Development of a Consensus Case Definition for Chronic Multisymptom Illness in 1990-1991 Gulf War Veterans, Board on the Health of Select Populations, & Institute of Medicine. Chronic Multisymptom Illness in Gulf War Veterans: Case Definitions Reexamined. National Academies Press (US). 2014.
45. Fukuda K, Nisenbaum R, Stewart G, et al. Chronic multisymptom illness affecting Air Force veterans of the Gulf War. *JAMA* 1998;280:981–988.
46. Cherry N, Creed F, Silman A, et al. Health and exposures of United Kingdom Gulf War veterans. Part II: the relation of health to exposure. *Occup Environ Med* 2001;58:299–306.
47. Fricker RD, Reardon E, Spektor DM, et al. A review of the scientific literature as it pertains to Gulf War illnesses. In: *Volume 12: Pesticide Use During The* Gulf War*: A Survey of* Gulf War *Veterans*. RAND; 2000.
48. Steele L, Quaden R, Ahmed ST, et al. Association of deployment characteristics and exposures with persistent ill health among 1990–1991 Gulf War veterans in the VA Million Veteran Program. *Environ Health* 2024;23:92.
49. Tarolli CG, Biernot JM, Creigh PD, et al. Practicing in a pandemic: a clinician's guide to remote neurologic care. *Neurol Clin Pract* 2021;11:e179–e188.
50. Wechsler LR, Tsao JW, Levine SR, et al. Teleneurology applications: report of the Telemedicine Work Group of the American Academy of Neurology. *Neurology* 2013;80:670–676.
51. Fischl B, Salat DH, Busa E, et al. Whole brain segmentation: automated labeling of neuroanatomical structures in the human brain. *Neuron* 2002;33:341–355.
52. Althouse AD. Adjust for multiple comparisons? It's not that simple. *Ann Thorac Surg* 2016;10:1644–1645.
53. Haley RW, Charuvastra E, Shell WE, et al. Cholinergic autonomic dysfunction in veterans with Gulf War illness: confirmation in a population-based sample. *JAMA Neurol* 2013;70:191–200.
54. Muller ML, Bohnen NI. Cholinergic dysfunction in Parkinson's disease. *Curr Neurol Neurosci Rep* 2013;13:377.
55. Dickey B, Madhu LN, Shetty AK. Gulf War illness: mechanisms underlying brain dysfunction and promising therapeutic strategies. *Pharmacol Ther* 2021;220:107716.
56. Dias V, Junn E, Mouradian MM. Role of oxidative stress in Parkinson's disease. *J Parkinsons Dis* 2013;3:461–491.
57. Garza-Lombo C, Thang M, Greve HJ, et al. Circulating HMGB1 is elevated in veterans with Gulf War illness and triggers the persistent pro-inflammatory microglia phenotype in male C57Bl/6J mice. *Transl Psychiatry* 2021;11:390.
58. Pajares M, I Rojo A, Manda G, Bosca L, Cuadrado A. Inflammation in Parkinson's disease: mechanisms and therapeutic implications. *Cells* 2020;9: 1687.
59. Haba-Rubio J, Frauscher B, Marques-Vidal P, et al. Prevalence and determinants of rapid eye movement sleep behavior disorder in the general population. *Sleep* 2018;41:zsx197.
60. Lee WJ, Baek SH, Im HJ, et al. REM sleep behavior disorder and its possible prodromes in general population: prevalence, polysomnography findings, and associated factors. *Neurology* 2023;101:e2364–e2375.
61. Elliott JE, Opel RA, Pleshakov D, et al. Posttraumatic stress disorder increases the odds of REM sleep behavior disorder and other parasomnias in veterans with and without comorbid traumatic brain injury. *Sleep* 2020;43:zsz237.
62. Barone DA. Trauma-associated sleep disorder. *Sleep Med Clin* 2024;19:93–99.
63. Mano M, Nomura A, Sasanabe R. Gender difference in REM sleep behavior disorder in Japanese population: Polysomnography and Sleep Questionnaire Study. *J Clin Med* 2024;13:914.
64. Husain AM, Miller PP, Carwile ST. Rem sleep behavior disorder: potential relationship to post-traumatic stress disorder. *J Clin Neurophysiol* 2001;18: 148–157.
65. Verma A, Anand V, Verma NP. Sleep disorders in chronic traumatic brain injury. *J Clin Sleep Med* 2007;3:357–362.
66. Sullivan K, Krengel M, Bradford W, et al. Neuropsychological functioning in military pesticide applicators from the Gulf War: effects on information processing speed, attention and visual memory. *Neurotoxicol Teratol* 2018; 65:1–13.
67. Krengel M, Sullivan K, Zundel CG, et al. Toxicant exposures and health symptoms in military pesticide applicators from the 1991 Gulf War. *J Occup Environ Med* 2024;66:e584–e592.
68. Hanyu H, Inoue Y, Sakurai H, et al. Voxel-based magnetic resonance imaging study of structural brain changes in patients with idiopathic REM sleep behavior disorder. *Parkinsonism Relat Disord* 2012;18:136–139.
69. Li G, Chen Z, Zhou L, et al. Altered structure and functional connectivity of the central autonomic network in idiopathic rapid eye movement sleep behaviour disorder. *J Sleep Res* 2021;30:e13136.
70. Fraigne JJ, Torontali ZA, Snow MB, Peever JH. REM sleep at its core—circuits, neurotransmitters, and pathophysiology. *Front Neurol* 2015;6:123.
71. Valencia Garcia S, Brischoux F, Clement O, et al. Ventromedial medulla inhibitory neuron inactivation induces REM sleep without atonia and REM sleep behavior disorder. *Nat Commun* 2018;9:504.
72. Vijayakumari AA, Sakaie KE, Fernandez HH, Walter BL. Parkinson's disease subtypes and their association with probable rapid eye movement sleep behavior disorder severity: a brainstem tractography and machine learning investigation. *Brain Imaging Behav* 2025;19:189–194.
73. García-Lorenzo D, Longo-Dos Santos C, Ewenczyk C, et al. The coeruleus/subcoeruleus complex in rapid eye movement sleep behaviour disorders in Parkinson's disease. *Brain* 2013;136(Pt 7):2120–2129.
74. Unger MM, Belke M, Menzler K, et al. Diffusion tensor imaging in idiopathic REM sleep behavior disorder reveals microstructural changes in the brainstem, substantia nigra, olfactory region, and other brain regions. *Sleep* 2010;33: 767–773.
75. Blanchard M, Molina-Vicenty HD, Stein PK, et al. Medical correlates of chronic multisymptom illness in Gulf War veterans. *Am J Med* 2019;132: 510–518.
76. Avery TJ, Mathersul DC, Schulz-Heik RJ, Mahoney L, Bayley PJ. Self-reported autonomic dysregulation in Gulf War illness. *Mil Med* 2023;188(7–8):usab546.
77. Fox A, Helmer D, Tseng CL, McCarron K, Satcher S, Osinubi O. Autonomic symptoms in Gulf War veterans evaluated at the war related illness and injury study center. *Mil Med* 2019;184(3–4):e191–e196.
78. McCarter SJ, Gehrking TL, St Louis EK, et al. Autonomic dysfunction and phenoconversion in idiopathic REM sleep behavior disorder. *Clin Auton Res* 2020;30:207–213.
79. Chen Z, Li G, Liu J. Autonomic dysfunction in Parkinson's disease: implications for pathophysiology, diagnosis, and treatment. *Neurobiol Dis* 2020;134:104700.
80. Tubbs AS, Dollish HK, Fernandez F, Grandner MA. The basics of sleep physiology and behavior. In: Grander MA, ed. *Sleep and Health*. Academic Press; 2019:3–10.
81. Research Advisory Committee on Gulf War Veterans' Illnesses. Gulf War Illness and the Health of Gulf War Veterans: Scientific Findings and Recommendations. U.S. Government Printing Office 2008–560–653. Washington, D.C. 2008. https://www.va.gov/RAC-GWVI/docs/Committee_Documents/GWIandHealthofGWVeterans_RAC-GWVIReport_2008.pdf. Accessed October 20, 2025.
82. Haley RW, Kurt TL, Hom J. Is there a Gulf War syndrome? *JAMA* 1997;277: 215–222.
83. Kim H, Jung HR, Kim JB, Kim DJ. Autonomic dysfunction in sleep disorders: from neurobiological basis to potential therapeutic approaches. *J Clin Neurol* 2022;18:140–151.
84. Duong LM, Nono Djotsa ABS, Vahey J, et al. Association of Gulf War illness with characteristics in deployed vs. non-deployed Gulf War era veterans in the

cooperative studies program 2006/million veteran program 029 cohort: a cross-sectional analysis. *Int J Environ Res Public Health* 2022;20:258.
85. Gifford EJ, Vahey J, Hauser ER, et al. Gulf War illness in the Gulf War Era Cohort and Biorepository: the Kansas and Centers for Disease Control definitions. *Life Sci* 2021;278:119454.
86. Haley RW, Dever JA, Kramer G, Teiber JF. The effect of disease misclassification on the ability to detect a gene-environment interaction: implications of the specificity of case definitions for research on Gulf War illness. *BMC Med Res Methodol* 2023;23:273.
87. Zundel CG, Heeren T, Grasso CM, et al. Changes in health status in the Ft. Devens Gulf War veterans cohort: 1997–2017. *Neurosci Insights* 2020;15: 2633105520952675.
88. Dursa EK, Cao G, Proter B, Culpepper WJ, Schneiderman AI. The health of Gulf War and Gulf Era Veterans over time: U.S. Department of Veterans Affairs' Gulf War Longitudinal Study. *J Occup Environ Med* 2021;63: 889–894.
89. Sullivan K, Krengel M, Heboyan V, et al. Prevalence and patterns of symptoms among female veterans of the 1991 Gulf War era: 25 years later. *J Womens Health (Larchmt)* 2020;29:819–826.
90. Barth SK, Kang HK, Bullman TA, Wallin MT. Neurological mortality among U.S. veterans of the Persian Gulf War: 13-year follow-up. *Am J Ind Med* 2009; 52:663–670.
91. Bullman TA, Mahan CM, Kang HK, Page WF. Mortality in US Army Gulf War veterans exposed to 1991 Khamisiyah chemical munitions destruction. *Am J Public Health* 2005;95:1382–1388.
92. Pang SYY, Ho PWL, Liu HF, et al. The interplay of aging, genetics and environmental factors in the pathogenesis of Parkinson's disease. *Transl Neurodegener* 2019;8:23.
93. Frigerio R, Elbaz A, Sanft KR, et al. Education and occupations preceding Parkinson disease: a population-based case-control study. *Neurology* 2005; 65:1575–1583.
94. Park J, Yoo CI, Sim CS, et al. Occupations and Parkinson's disease: a multi-center case-control study in South Korea. *Neurotoxicology* 2005;26:99–105.
95. Wright Willis A, Evanoff BA, Lian M, Criswell SR, Racette BA. Geographic and ethnic variation in Parkinson disease: a population-based study of US Medicare beneficiaries. *Neuroepidemiology* 2010;34:143–151.
96. Van Den Eeden SK. Incidence of Parkinson's disease: variation by age, gender, and race/ethnicity. *Am J Epidemiol* 2003;157:1015–1022.
97. Llibre-Guerra JJ, Prina M, Sosa AL, et al. Prevalence of parkinsonism and Parkinson disease in urban and rural populations from Latin America: a community based study. *Lancet Reg Health Am* 2022;7:100136.
98. Mayeux R, Marder K, Cote LJ, et al. The frequency of idiopathic Parkinson's disease by age, ethnic group, and sex in Northern Manhattan, 1988–1993. *Am J Epidemiol* 1995;142:820–827.
99. Duarte Folle A, Flores MES, Kusters C, et al. Ethnicity and Parkinson's disease: motor and nonmotor features and disease progression in Latino patients living in rural California. *J Gerontol A Biol Sci Med Sci* 2023;78: 1258–1268.
100. Marrero-Gonzalez P, Iranzo A, Bedoya D, et al. Prodromal Parkinson disease in patients with idiopathic hyposmia. *J Neurol* 2020;267:3673–3682.
101. Stiasny-Kolster K, Doerr Y, Moller JC, et al. Combination of 'idiopathic' REM sleep behaviour disorder and olfactory dysfunction as possible indicator for alpha-synucleinopathy demonstrated by dopamine transporter FP-CIT-SPECT. *Brain* 2005;128(Pt 1):126–137.
102. Postuma RB, Lang AE, Massicotte-Marquez J, Montplaisir J. Potential early markers of Parkinson disease in idiopathic REM sleep behavior disorder. *Neurology* 2006;66:845–851.
103. Orso B, Brosse S, Frasnelli J, Arnaldi D. Opportunities and pitfalls of REM sleep behavior disorder and olfactory dysfunction as early markers in Parkinson's disease. *J Parkinsons Dis* 2024;14(Suppl 2):S275–S285.
104. Doty RL. Neurotoxic exposure and impairment of the chemical senses of taste and smell. *Handb Clin Neurol* 2015;131:299–324.
105. Schwartz BS, Doty RL, Monroe C, Frye R, Barker S. Olfactory function in chemical workers exposed to acrylate and methacrylate vapors. *Am J Public Health* 1989;79:613–618.
106. Nag N, Jelinek GA. More research is needed on lifestyle behaviors that influence progression of Parkinson's disease. *Front Neurol* 2019;10:452.
107. Fang X, Han D, Cheng Q, et al. Association of levels of physical activity with risk of Parkinson disease: a systematic review and meta-analysis. *JAMA Netw Open* 2018;1:e182421.
108. Muller J, Myers J. Association between physical fitness, cardiovascular risk factors, and Parkinson's disease. *Eur J Prev Cardiol* 2018;25:1409–1415.
109. Ross GW, Abbott RD, Petrovitch H, et al. Association of coffee and caffeine intake with the risk of Parkinson disease. *JAMA* 2000;283:2674–2679.
110. Bisaglia M. Mediterranean diet and Parkinson's disease. *Int J Mol Sci* 2022;24:42.
111. Balakrishnan R, Azam S, Cho DY, Su-Kim I, Choi DK. Natural phytochemicals as novel therapeutic strategies to prevent and treat Parkinson's disease: current knowledge and future perspectives. *Oxid Med Cell Longev* 2021;2021: 6680935.
112. Yoon SY, Park YH, Lee HJ, Kang DR, Kim YW. Lifestyle factors and Parkinson disease risk Korean nationwide cohort study with repeated health screening data. *Neurology* 2021;98:e641–e652.